Year 9 Topics in exam

Electromagnetism.

- Poles of a magnet
- Permanent and Induced Magnetism
- Magnetic Fields
    - Bar Magnet
    - Attracting/Repelling
    - Current Carrying Wire
    - Solenoid
- Electromagnetism
- Electromagnetism uses e.g Bell, Relay, loudspeaker

AQA Kerboodle textbook & Revision guide relevant chapters:

P15.1, 15.2, 15.3

Waves

- Wave properties (wavespeed = frequency x wavelength) (period = 1 / frequency)
- Reflection of light
- Refraction of light
- Speed of light/ nature e.g.travel through a vacuum, transverse
- Dispersion by a prism
- Colour/uses of colour filters
- Electromagnetic spectrum
- Sound waves/nature e.g. speed/do not travel through a vacuum/caused by vibrations
- Frequency and pitch/loudness and amplitude and CRO traces
- Ultrasound and Seismic waves

AQA Kerboodle textbook & Revision guide relevant chapters:

All of P12 except 12.2, all of P13 and P14.1,P14.2 & 14.3

Mechanics 1

- Speed and velocity $v=s/t$
- Acceleration (a=change in velocity/time $a=\Delta v/t$)
- Distance time and velocity time graphs
- Newton's first and second laws of motion $F=Ma$
- Weight $W=mg$
- Terminal velocity

AQA Kerboodle textbook & Revision guide relevant chapters:

P8.1,8.2 & 8.3. P9 all chapters, P10.1 & 10.2

Equations to learn:

$Time\ period = \frac{1}{Frequency}$ $T = \frac{1}{f}$

$Wave\ speed = frequency\ x\ wavelength$ $v = f\lambda$

$Weight = mass\ x\ strength\ of\ gravity$ $W = mg$

$Distance = average\ speed\ x\ time$ $s = vt$

$Acceleration = \frac{Change\ in\ velocity}{Time\ taken}$ $a = \frac{\Delta v}{t}$ or $a = \frac{v-u}{t}$

$Resultant\ force = mass \times acceleration$ $F=ma$